

LUCAS (HUNG CHUN) HSU

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RESEARCH INTEREST

To develop **Language-Agent Architectures** that enable **Multi-Agent Systems (MAS)** with multimodal world perception capabilities, scalable agent auto-routing, and multi-source context processing. My research focuses on 1) **Multimodal LLMs**, 2) **Agent Orchestration**, and 3) **Personalized Conversational Information Retrieval**. This vision originated from my early Graph Neural Networks research starting in 2021, where I conceptualized transforming node-to-node social networks into LLM-human networks—a trending research topic we now call **Human-Agent Collaboration Systems**.

EDUCATION

M.S. in Data Science (Thesis). National Taiwan University

Sep 2021 - Aug 2023

- Joint Master's Program with Academia Sinica, National Research Academy of Taiwan
- Graduated 1% of the class (GPA: 4.0/4.0) with Academic Excellence Scholarship and Phi-Tau-Phi Scholastic Honor Membership

B.S. in Engineering and System Science. National Tsing Hua University

Sep 2017 - Jan 2021

- Specialized in Applied Mathematics and Advanced Statistics for Microelectronic Engineering

PUBLICATIONS

LODESTAR: Trustworthy Entropy Is Navigated, Not Merely Measured

Hung-Chun Hsu, Po-Jen Ko, Che-Cheng Wu, Li-Yang Chang, Chuan-Ju Wang

arXiv preprint

Test-Time Scaling Strategies for LLM-based Generative Retrieval in Multimodal Conversational Recommendations

Hung-Chun Hsu, Yuan-Ching Kuo, Chao-Han Huck Yang, Szu-Wei Fu, Hanrong Ye, Hongxu Yin,

Yu-Chiang Frank Wang, Ming-Feng Tsai, Chuan-Ju Wang

arXiv preprint

CFDA@TREC iKAT 2025: Enhancing Personalized Conversational Search through Query Reformulation and Rank Fusion

Yu-Cheng Chang, Guan-Wei Yeo, Quah Eugene, Fan-Jie Shih, Yuan-Ching Kuo, Tsung-En Yu,

Hung-Chun Hsu, Ming-Feng Tsai, Chuan-Ju Wang

NIST TREC 2025

FincGAN: A GAN Framework of Imbalanced Node Classification on Heterogeneous Graph Neural Network

Hung-Chun Hsu, Ting-Le Lin*, Bo-Jun Wu, Ming-Yi Hong, Che Lin, and Chih-Yu Wang*

ICASSP 2024

RESEARCH EXPERIENCE

Research Center for Information Technology Innovation, Academia Sinica

Apr 2024 - Present

Senior Research Assistant and Project Lead at the CFDA Lab

Taipei, Taiwan

- Leading the Information Retrieval (IR) research project in collaboration with [NVIDIA Research](#) (Taiwan and US) under the supervision of Prof. Chuan-Ju Wang and Prof. Ming-Feng Tsai. Focused on conversational retrieval and Multimodal-LLMs.
- Participated twice in NIST's Text Retrieval Conference (TREC) tracks starting from 2024: Product Recommendations and Interactive Knowledge Assistance (iKAT). Led and coordinated the competition teams with up to 6 direct reports.

Center of Digital, Data and Technology, Cathay Financial Holdings Co. Ltd.

Feb 2024 - Apr 2024

Data Scientist Intern - R&D

Taipei, Taiwan

- Developed an enterprise-level Chinese legal retrieval pipeline leveraging LangChain, increasing recall@50 performance from 23% to 39% compared to LangChain's basic two-staged retrieval settings through a novel legal articles pre-classification approach.

Research Center for Information Technology Innovation, Academia Sinica

Sep 2021 - Feb 2024

Full-time (2023 - 2024) and Part-time (2021 - 2023) Research Assistant at the SNAC Lab

Taipei, Taiwan

- Developed novel GAN-based data augmentation frameworks for heterogeneous graphs, addressing critical node class imbalance issues in graph neural networks.
- Achieved 14.4% improvement in F-score and 12.3% in PR-AUC over SOTA baseline GraphSMOTE through innovative handling of structured graph $G=(V, E, X)$. Published research in IEEE ICASSP with ongoing follow-up studies.

REPRESENTATIVE PROJECT

Advancing NVIDIA X-VILA via Constrained Decoding - with NVIDIA Research US

Sep 2025 - Present

- Enhance NVIDIA's multimodal LLM X-VILA with multimodal object retrieval capabilities (e.g., video retrieval) to ensure reliable multimodal information acquisition and eliminate multimodal hallucinations.

2025 TREC Interactive Knowledge Assistance Track (iKAT)

Jun 2025 - Sep 2025

- Led a team of 7 members to participate in the iKAT track, developing pipelines for traditional offline conversational search and exploring real-time conversational retrieval methods requiring low latency.
- Achieved 2nd place in Offline Passage Ranking and 3rd place in Online Interactive Generation among participating teams, outperforming the competition median by 10 pts in nDCG@3 and 11 pts in average human evaluation score respectively.

Multimodal Conversational Product Search with MLLMs - with NVIDIA Research Taiwan

Feb 2024 - July 2025

- Developed product recommendation system supporting any-to-any multimodal data using novel MLLM-based generative retrieval to predict consumer purchases, enhanced by our test-time reranking mechanism with average improvements of 14.5 points MRR and 10.6 points nDCG@1 across 3 benchmarks. Research findings are targeted for submission to **ACL ARR**.

2024 TREC Product Search and Recommendations Track

Jun 2024 - Sep 2024

- Led a 4-member team in the product search and recommendations track. Developed a multimodal product retrieval system leveraging BLIP, ViLT, and BEiT-3 as dual encoders and cross encoders.
- Implemented Weighted Sum Fusion to combine unimodal and multimodal retrieval results, achieving nDCG@10 of 76% compared to 72% from traditional BM25 and SPLADE retrievers.

ACADEMIC SERVICES

Complimentary AI Solution Consulting Services

AI Cooperative, Academia Sinica, Taiwan

- Provided AI technical consulting services to three domestic academic research teams and industry companies across diverse applications: (1) sea turtle blood leukocyte identification, (2) dolphin spectrogram recognition, (3) brain tumor cortical generation, and (4) stock market index prediction using language models for financial text analysis.

Academic Peer Review

- Served as reviewer for 21 papers submitted to conferences including ACL[25,26], AAI[25,26,27], NeurIPS[24,25,26], ICLR[24,25], SIGIR[26], and IJCAI[26].

TALKS

All talks presented at Joint Laboratory Seminar, Research Center of Information Technology Innovation, Academia Sinica, Taiwan

The Future of AI Interaction: True Multimodal Conversations

Sep 2025

- Delivered comprehensive overview of advanced 2025 multimodal retrieval methodologies, demonstrated novel cross-modality alignment loss functions for autoregressive models and proposed future directions in unified retrieval and generation MLLMs.

Generative Retrieval with FM-index and Ranking Objectives

Jan 2025

- Presented Generative Document Retrieval approaches with ranking-optimized objectives, analyzing FM-index implementation for semantic ID-based document representation to achieve improved search time complexity over traditional systems like FAISS.

Discrete Semantic Representations for Dense Passage Retrieval

Aug 2024

- Introduced approaches using learnable Semantic IDs for document retrieval, including DSI, VQ-VAE, RQ-VAE, and LMIndexer.

HONORS AND AWARDS

2023 Scholastic Honor Membership

awarded by The Phi-Tau-Phi Scholastic Honor Society of the R.O.C. (Taiwan)

2023 Graduation Ceremony Representative

awarded by the Data Science Master Program, National Taiwan University and Academia Sinica

2023 Academic Excellence Scholarship

awarded by National Taiwan University and Academia Sinica

2022 2nd Place in National Data Competition - Carrefour Taiwan E-commerce Next Purchase Prediction

awarded by Institute for Information Industry, Ministry of Digital Affairs, Executive Yuan of the R.O.C. (Taiwan)